# 17、添加用户NvM\Fee\Fls测试数据接口

卖家'漩涡遨游aa' 专属资料，非在漩涡遨游aa 购买资料，不提供答疑(一次购买仅提供有限次数10次答疑)

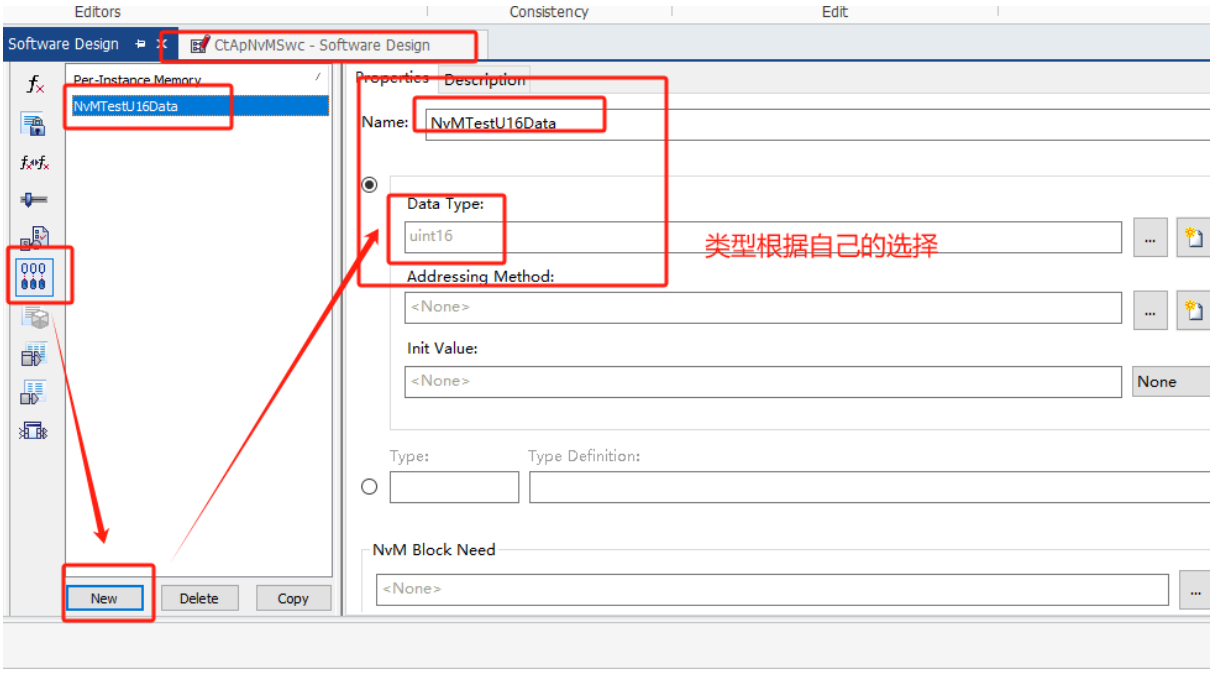
通过添加Swc，从Swc新建NvM 块开始，让Swc 可以通过Rte接口读写NvM数据，数据是支持掉电保存的，因为NvM最终调用的是Fls

## 一、新建CtApNvMSwc

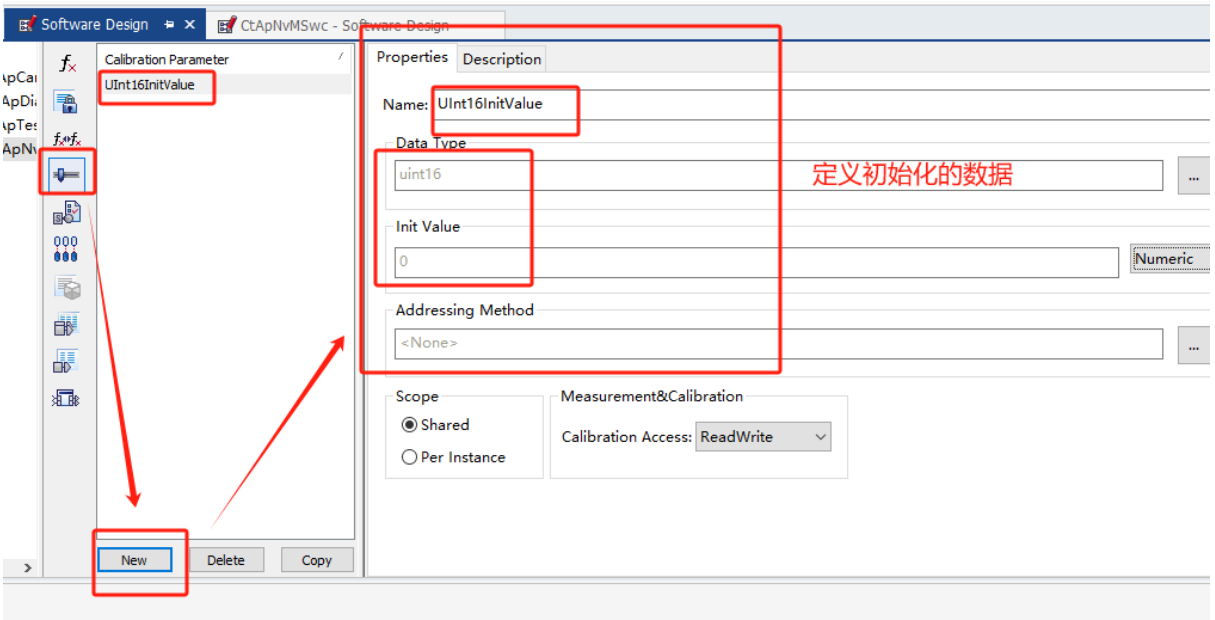
参考新建IO翻转的CtApTestSwc的方式添加Swc，同样添加init , 10ms ， 1000ms

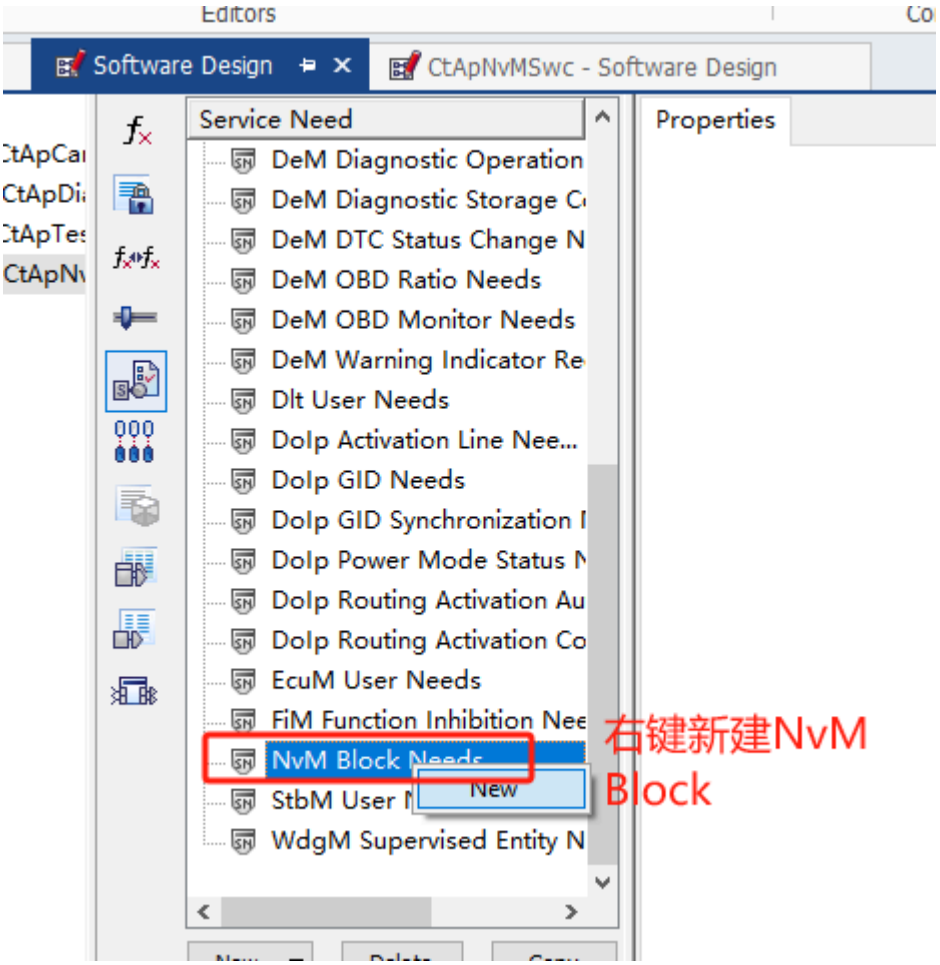
## 二、CtApNvMSwc新建NvM块相关数据

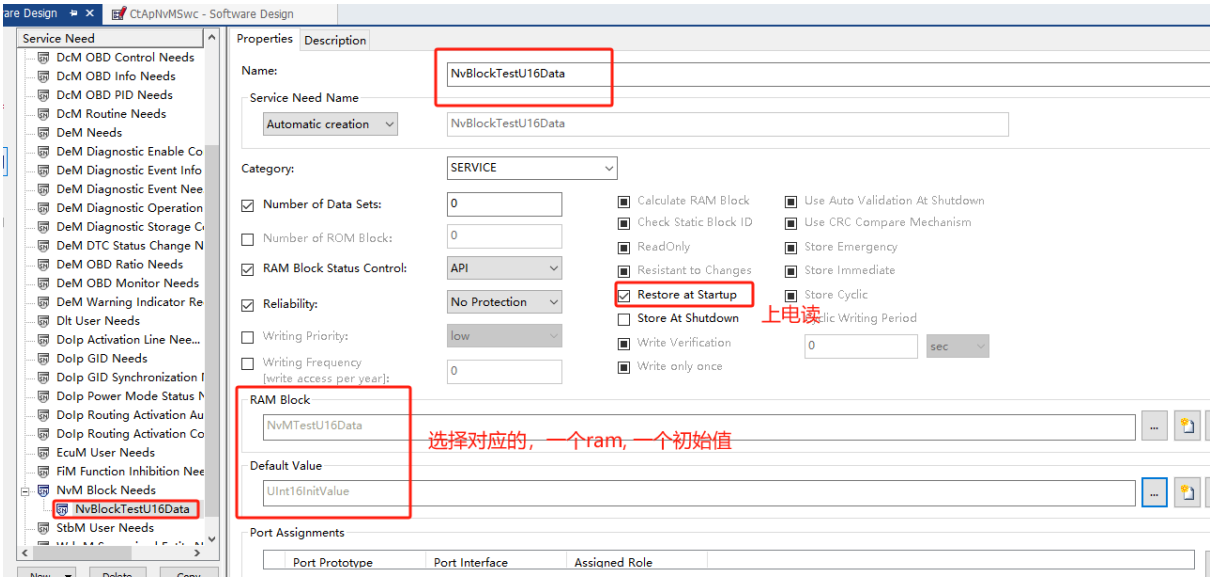
定义ram 缓存的数据，这个变量会在Rte.c 里面生成



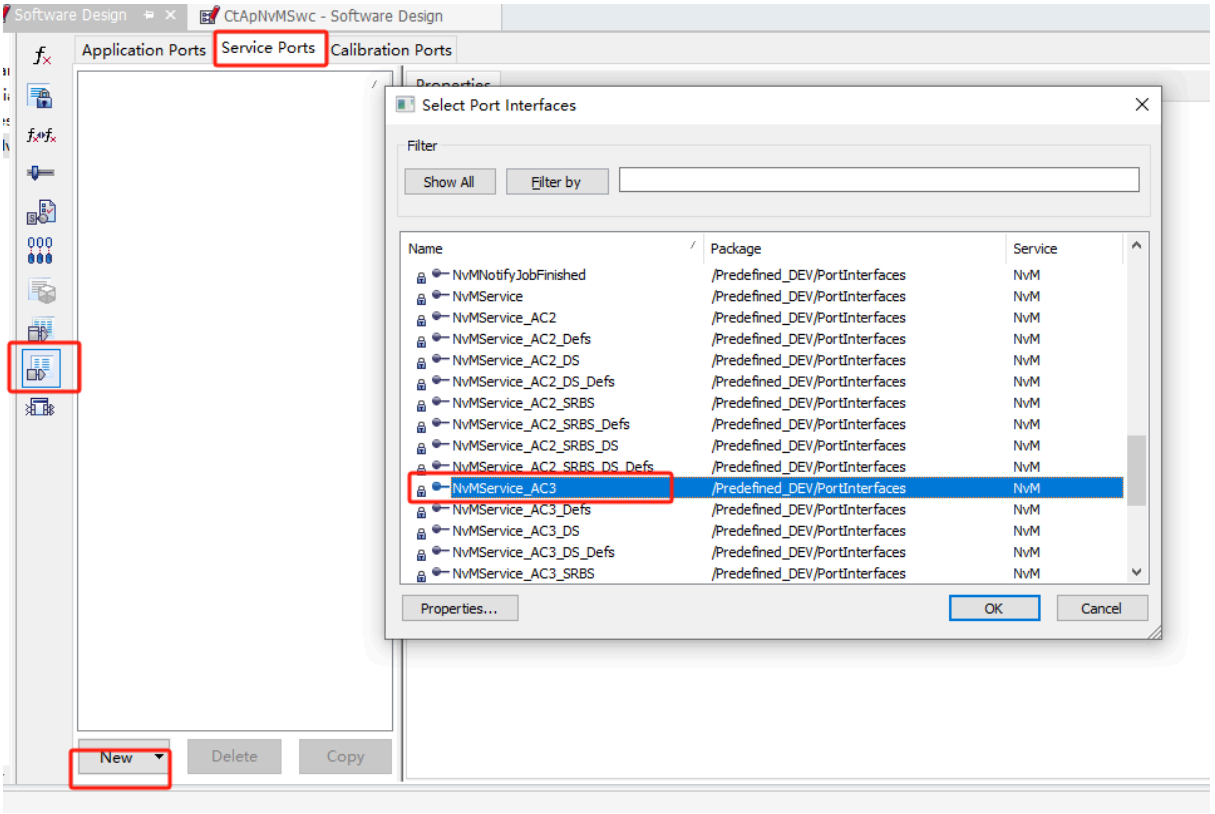

定义上面变量的初始化值为0

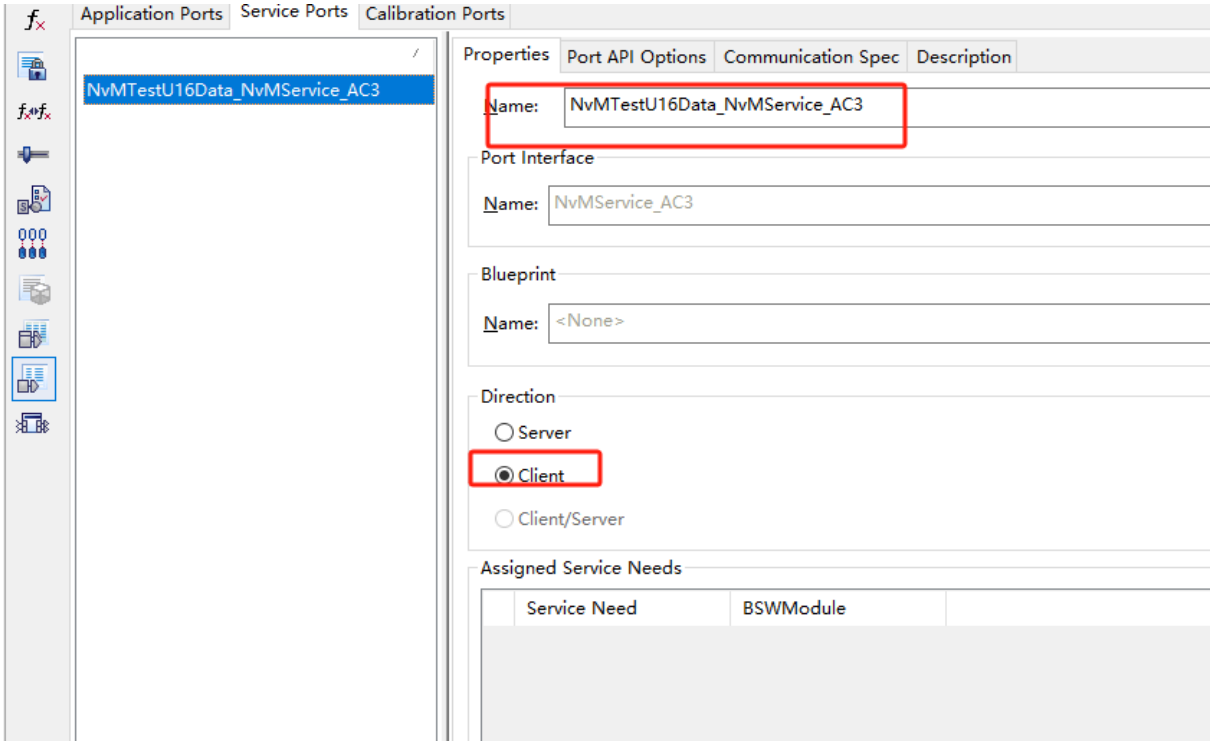


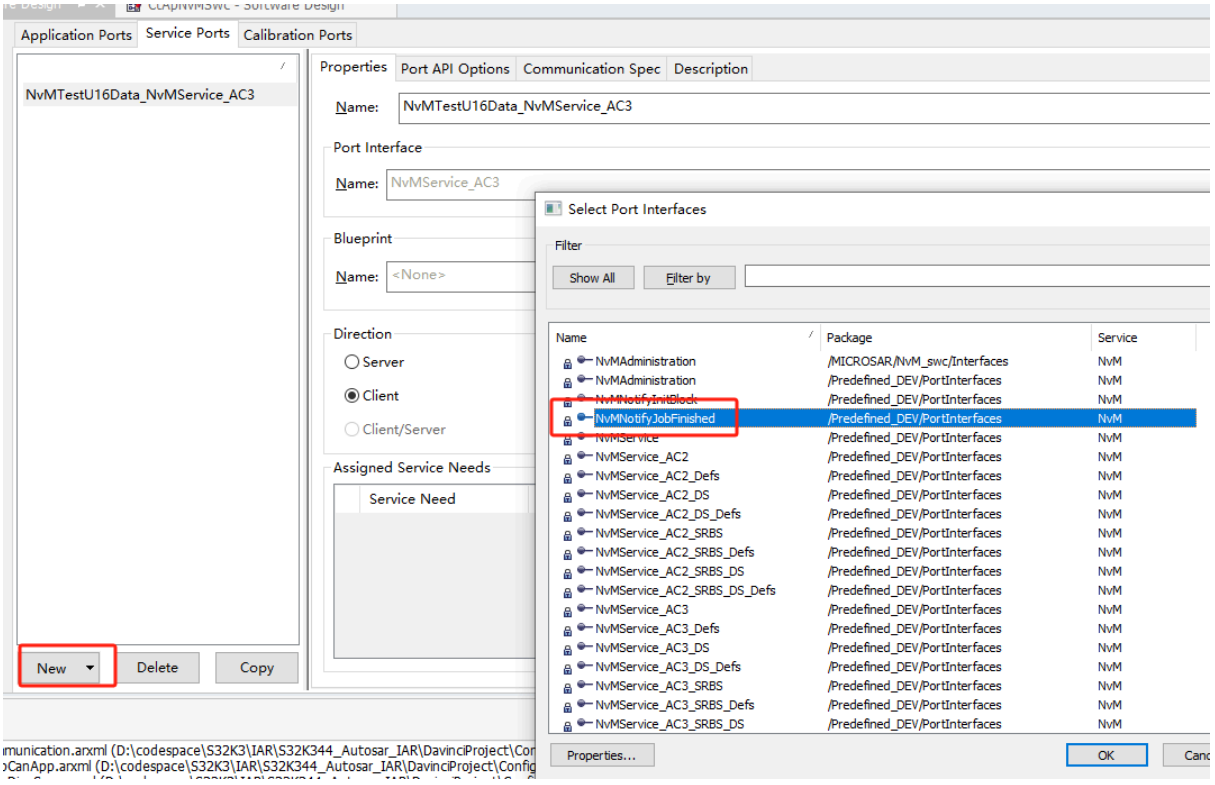


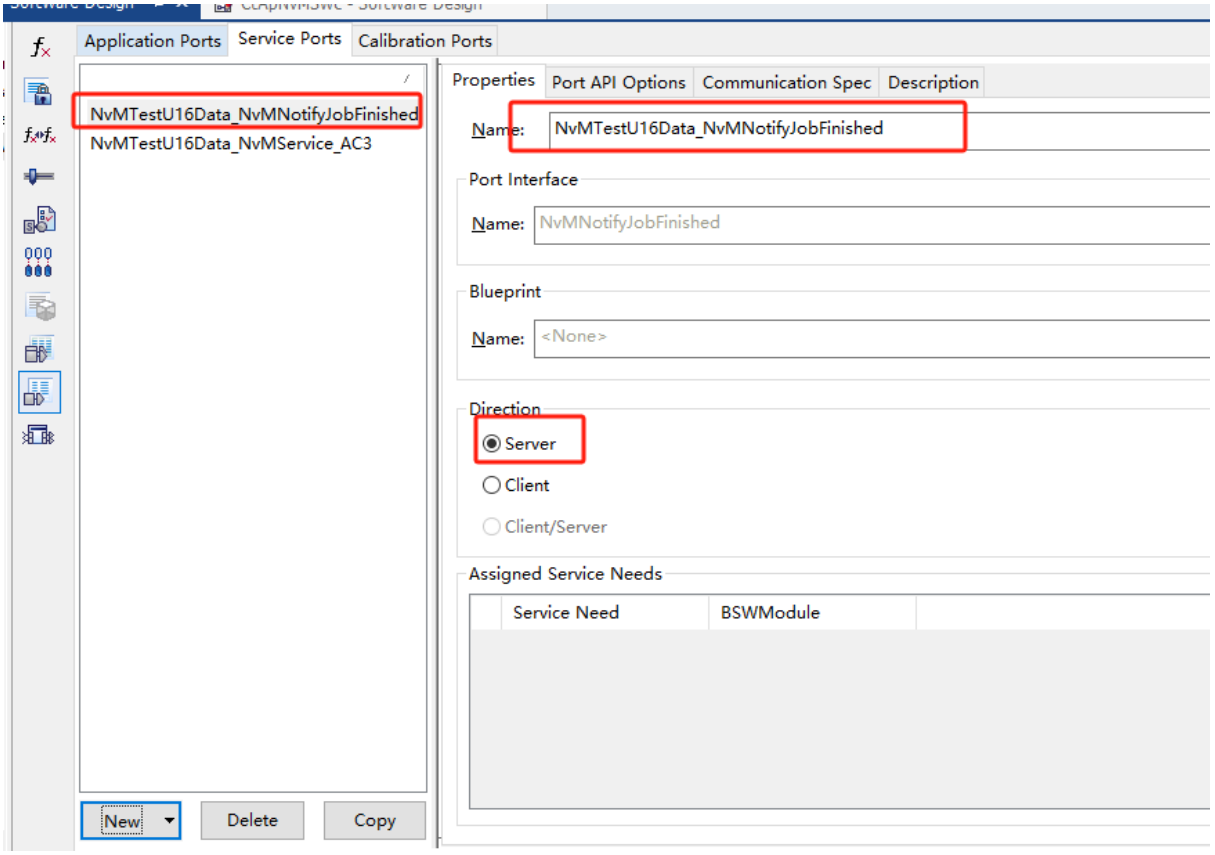



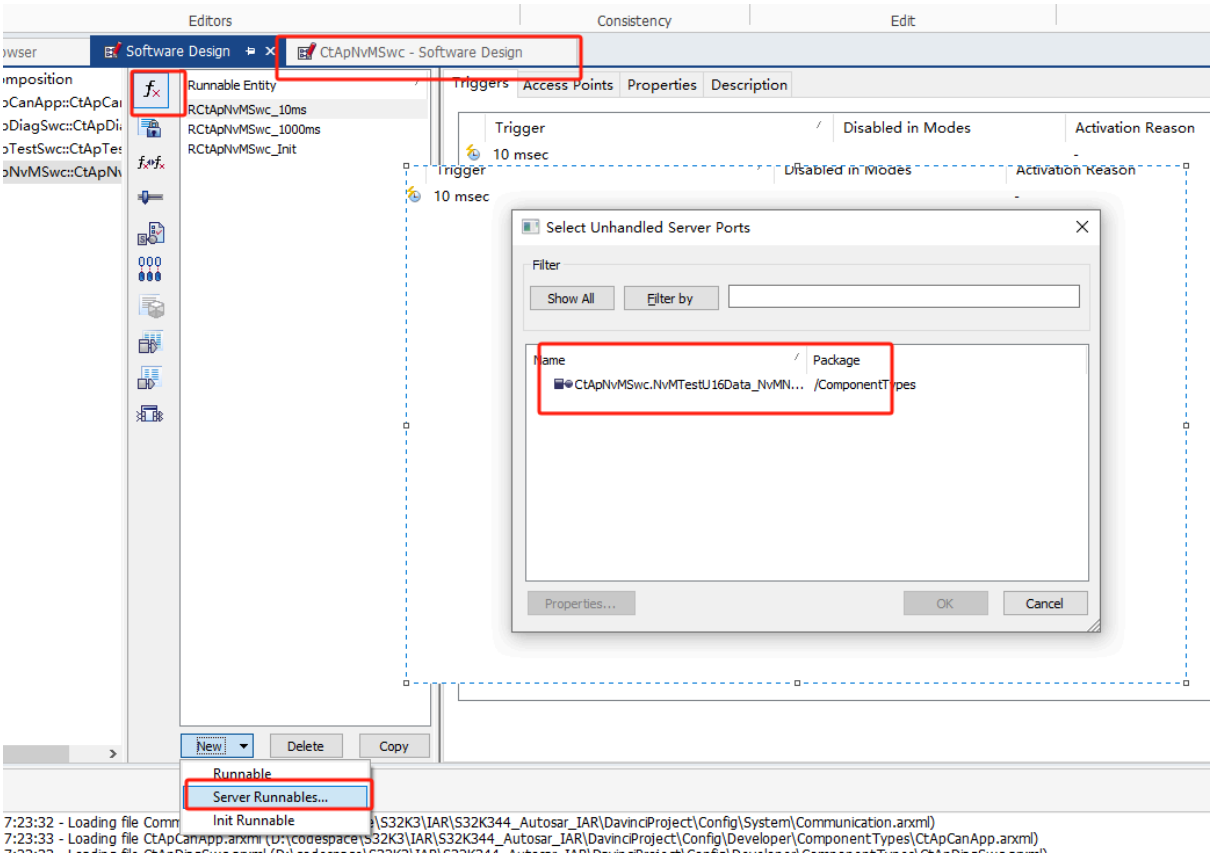

添加Rte的接口port

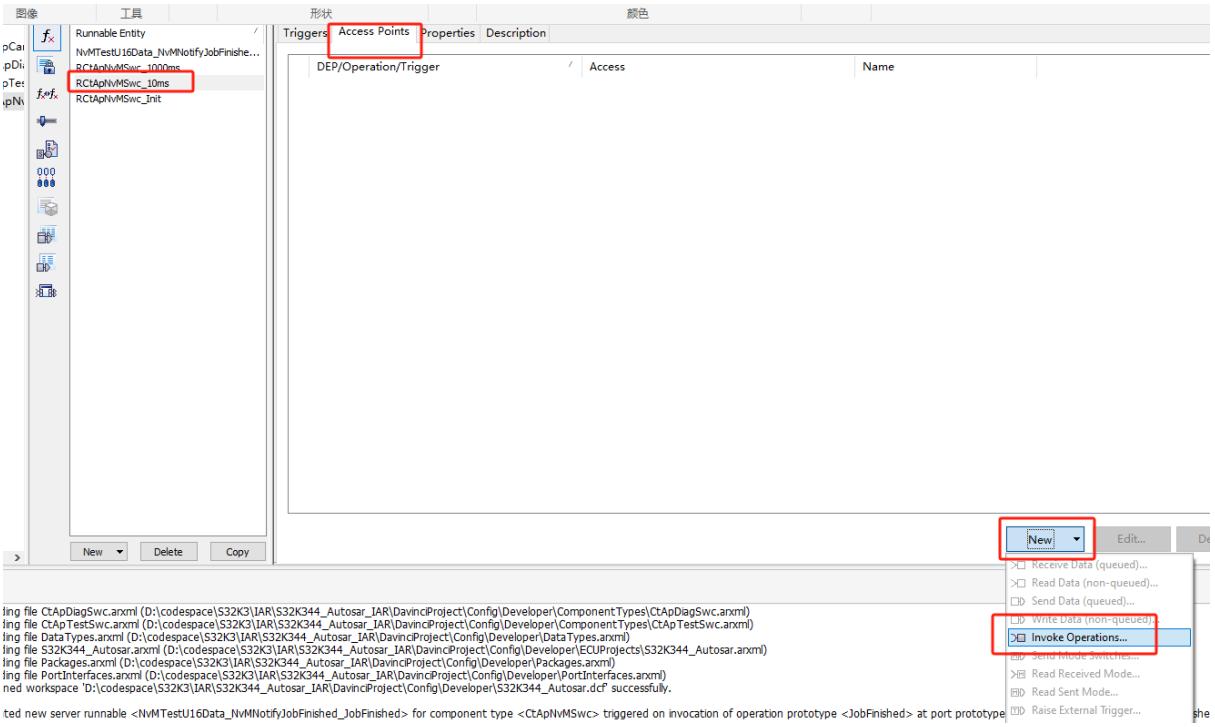

Software Design
CtApNvMSwc - Software Design
Application Ports
Service Ports
Calibration Ports
Select Port Interfaces
Filter
Show All
Filter by
Name
Package
Service
NvMNotifyJobFinished /Predefined_DEV/PortInterfaces NvM
NvMService /Predefined_DEV/PortInterfaces NvM
NvMService_AC2 /Predefined_DEV/PortInterfaces NvM
NvMService_AC2_Defs /Predefined_DEV/PortInterfaces NvM
NvMService_AC2_DS /Predefined_DEV/PortInterfaces NvM
NvMService_AC2_DS_Defs /Predefined_DEV/PortInterfaces NvM
NvMService_AC2_SRBS /Predefined_DEV/PortInterfaces NvM
NvMService_AC2_SRBS_Defs /Predefined_DEV/PortInterfaces NvM
NvMService_AC2_SRBS_DS /Predefined_DEV/PortInterfaces NvM
NvMService_AC2_SRBS_DS_Defs /Predefined_DEV/PortInterfaces NvM
NvMService_AC3 /Predefined_DEV/PortInterfaces NvM
NvMService_AC3_Defs /Predefined_DEV/PortInterfaces NvM
NvMService_AC3_DS /Predefined_DEV/PortInterfaces NvM
NvMService_AC3_DS_Defs /Predefined_DEV/PortInterfaces NvM
NvMService_AC3_SRBS /Predefined_DEV/PortInterfaces NvM
Properties...
OK
Cancel
New
Delete
Copy

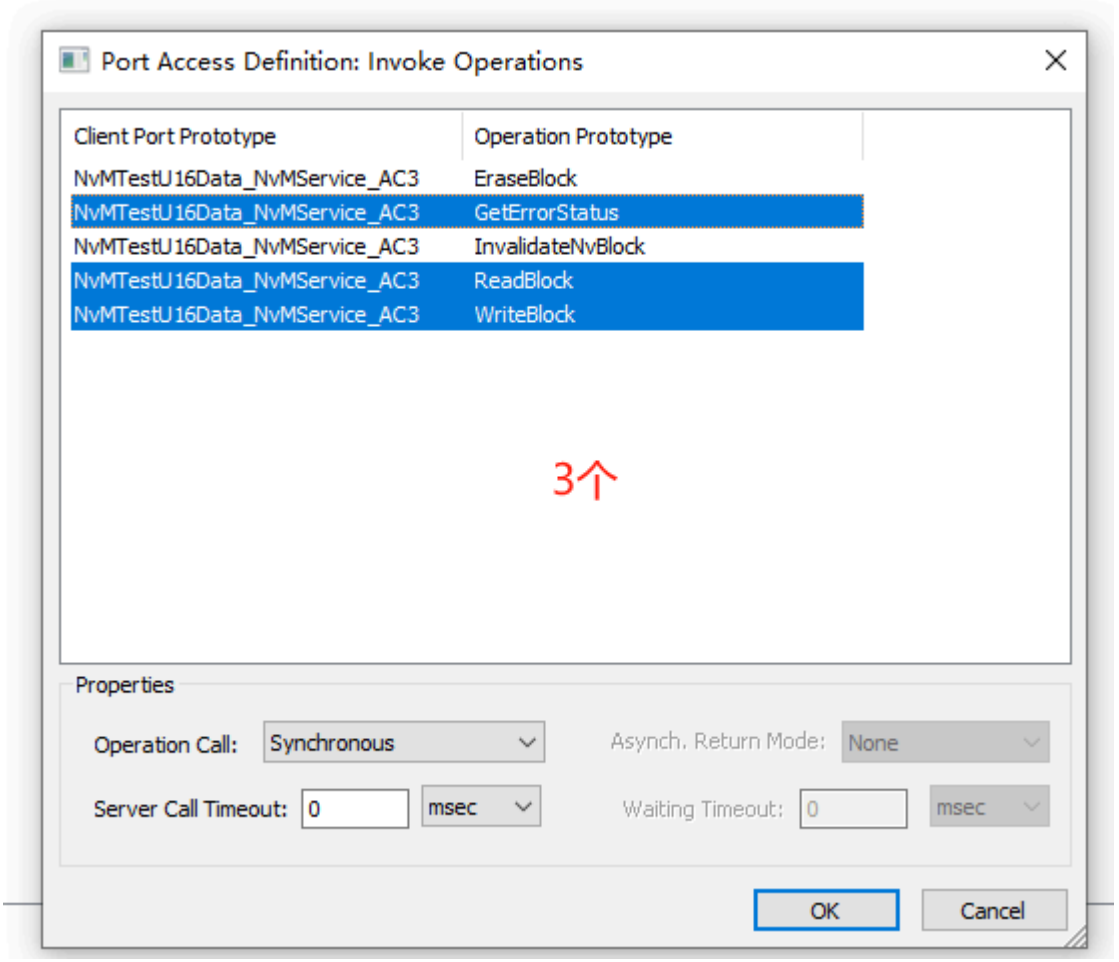

Application Ports
Service Ports
Calibration Ports
NvMTestU16Data_NvMService_AC3
Properties
Port API Options
Communication Spec
Description
Name: NvMTestU16Data_NvMService_AC3
Port Interface
Name: NvMService_AC3
Blueprint
Name: <None>
Direction
Server
Client
Client/Server
Assigned Service Needs
Service Need
BSWModule

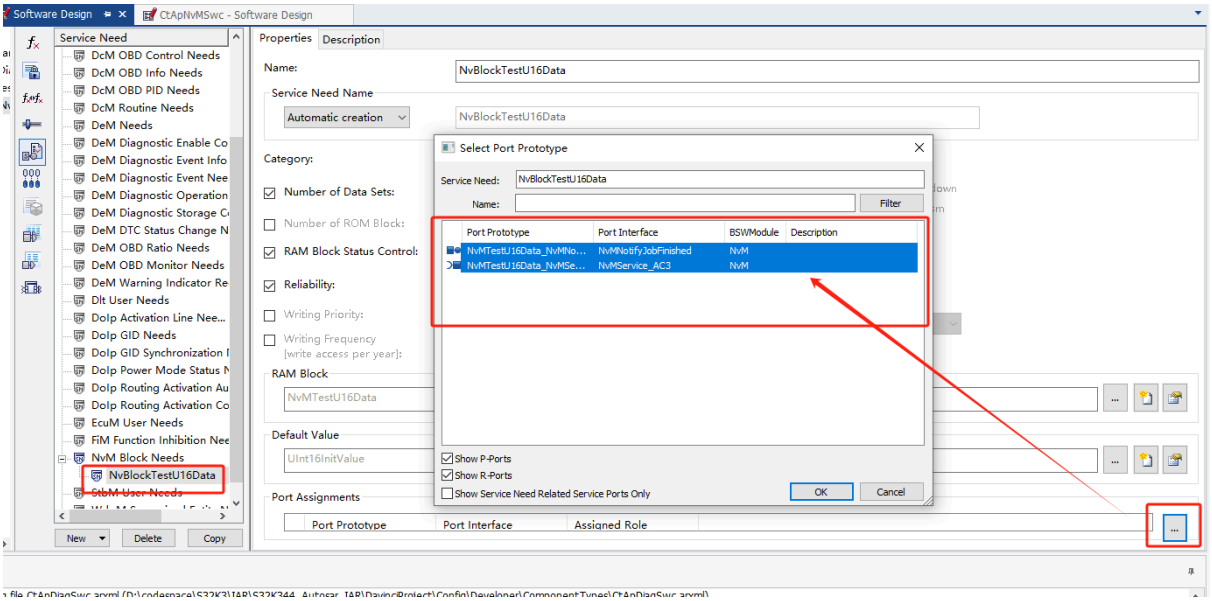

Application Ports
Service Ports
Calibration Ports
NvMTestU16Data_NvMService_AC3
Properties
Port API Options
Communication Spec
Description
Name:
NvMTestU16Data_NvMService_AC3
Port Interface
Name:
NvMService_AC3
Blueprint
Name:
<None>
Direction
Server
Client
Client/Server
Assigned Service Needs
Service Need
New
Delete
Copy
Select Port Interfaces
Filter
Show All
Filter by
Name
Package
Service
NvMAdministration
/MICROSAR/NvM_swc/Interfaces
NvMAdministration
/Predefined_DEV/PortInterfaces
NvMNotifyInitBlock
NvMNotifyJobFinished
NvMService
NvMService_AC2
NvMService_AC2_Defs
NvMService_AC2_DS
NvMService_AC2_DS_Defs
NvMService_AC2_SRBS
NvMService_AC2_SRBS_Defs
NvMService_AC2_SRBS_DS
NvMService_AC2_SRBS_DS_Defs
NvMService_AC3
NvMService_AC3_Defs
NvMService_AC3_DS
NvMService_AC3_DS_Defs
NvMService_AC3_SRBS
NvMService_AC3_SRBS_Defs
NvMService_AC3_SRBS_DS
NvM
Properties...
OK


Application Ports
Service Ports
Calibration Ports
NvMTestU16Data_NvMNotifyJobFinished
NvMTestU16Data_NvMService_AC3
Properties
Port API Options
Communication Spec
Description
Name:
NvMTestU16Data_NvMNotifyJobFinished
Port Interface
Name:
NvMNotifyJobFinished
Blueprint
Name:
<None>
Direction
Server
Client
Client/Server
Assigned Service Needs
Service Need
BSWModule
New
Delete
Copy

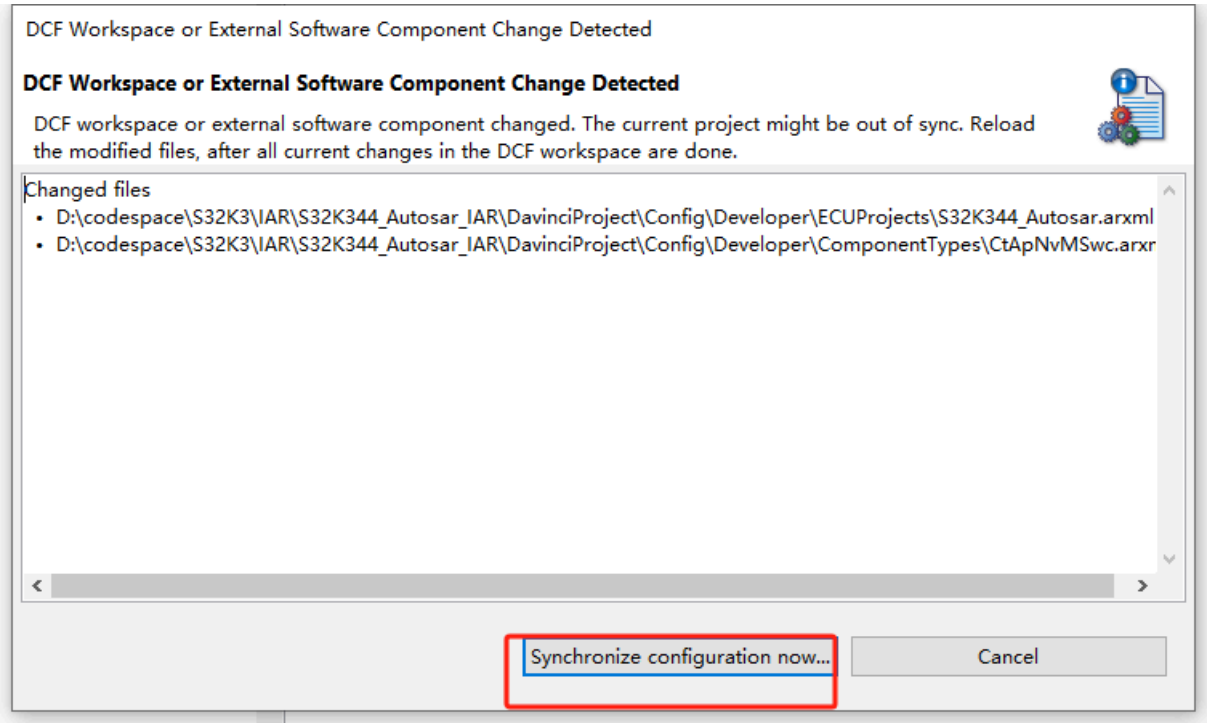

Editors
Consistency
Edit
Software Design
CtApNvMSwc - Software Design
Runnable Entity
RCtApNvMSwc_10ms
RCtApNvMSwc_1000ms
RCtApNvMSwc_Init
Triggers
Access Points
Properties
Description
Trigger
Disabled in Modes
Activation Reason
10 msec
Select Unhandled Server Ports
Filter
Show All
Filter by
Name
Package
CtApNvMSwc.NvMTestU16Data_NvMN...
/ComponentTypes
Properties...
OK
Cancel
New
Delete
Copy
Runnable
Server Runnables...
Init Runnable


Runnable Entity
NvMTestU16Data_NvMNotifyJobFinishe...
RCtApNvMSwc_1000ms
RCtApNvMSwc_10ms
RCtApNvMSwc_Init
Triggers
Access Points
Properties
Description
DEP/Operation/Trigger
Access
Name
New
Delete
Copy
Edit...
Receive Data (queued)...
Read Data (non-queued)...
Send Data (queued)...
Write Data (non-queued)...
Invoke Operations...
Read Received Mode...
Read Sent Mode...
Raise External Trigger...



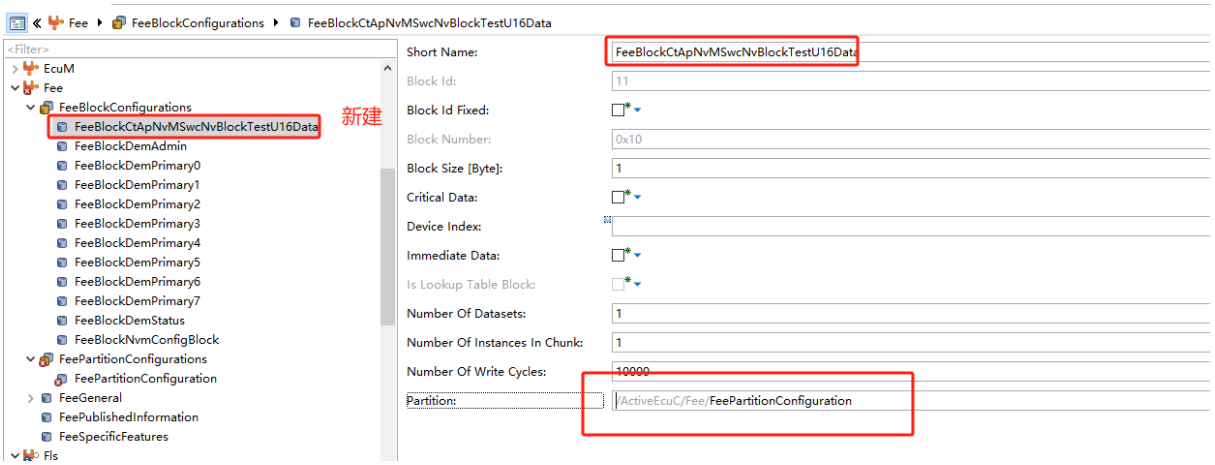



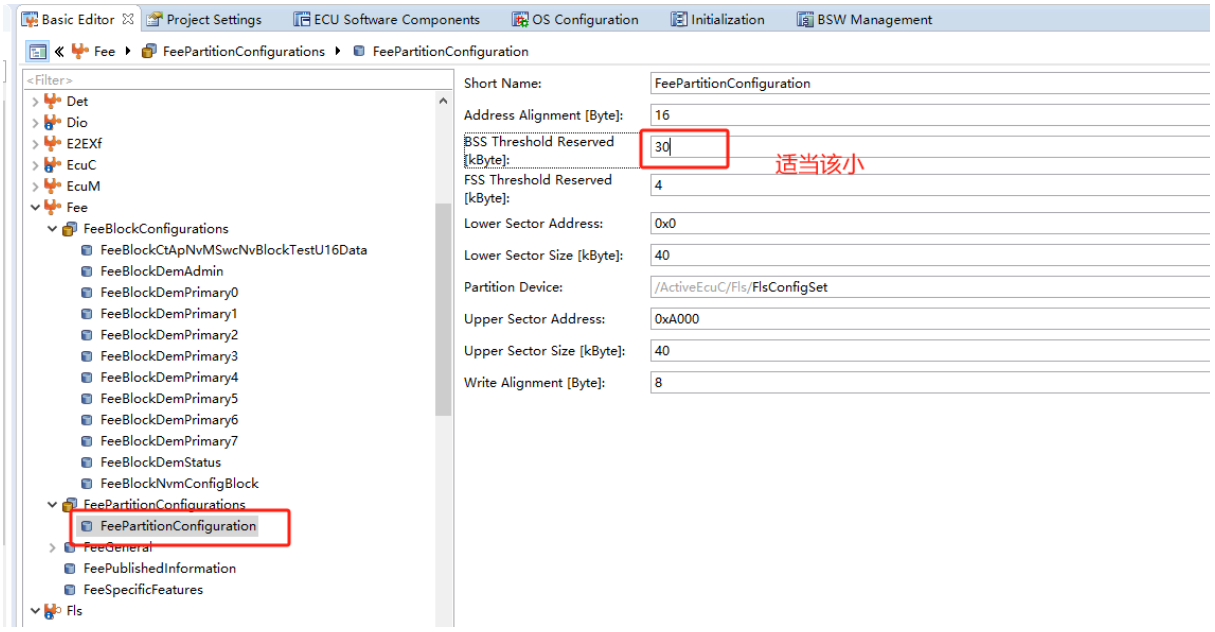

最后保存同步到davinci




# 三、cfg 添加Fee, 修改NvM

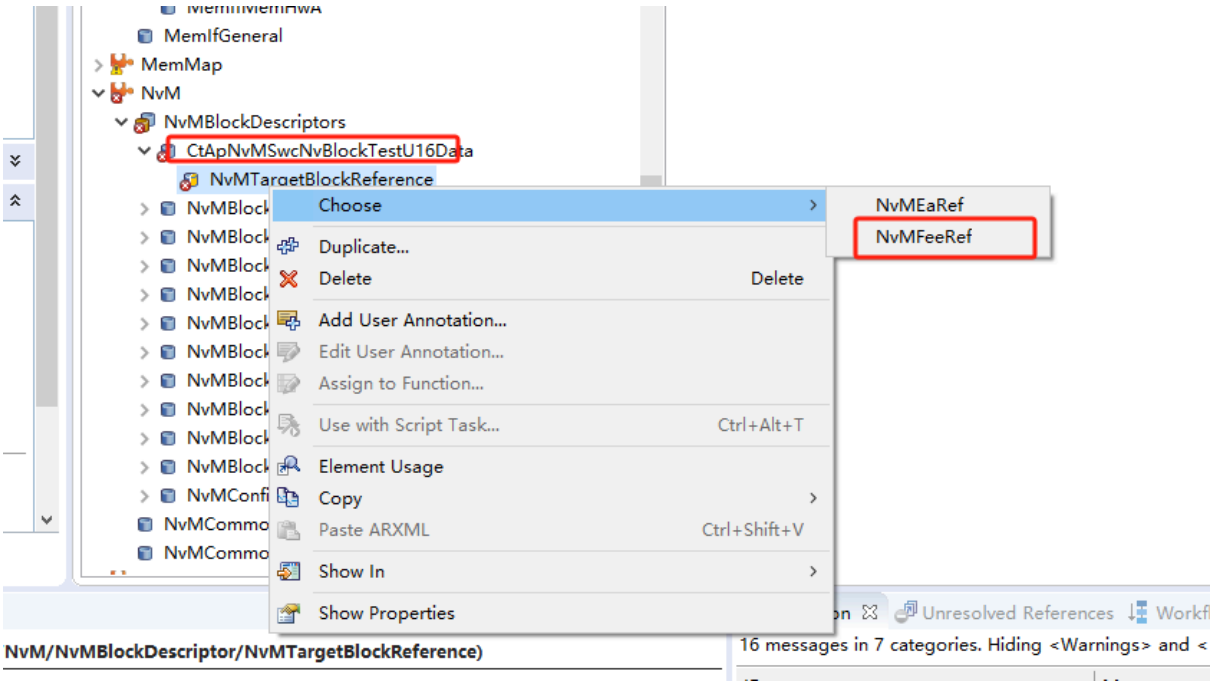
## 3.1、添加FeeBlock

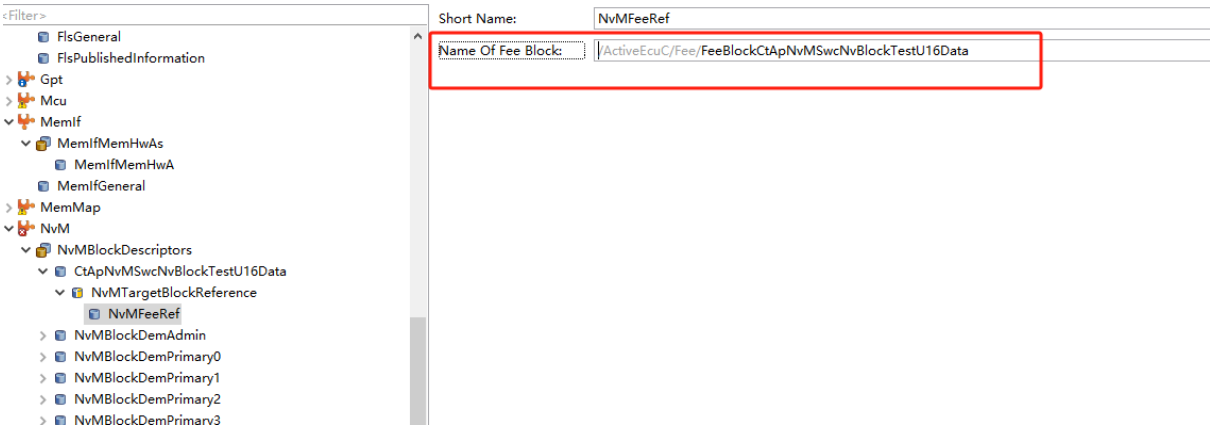


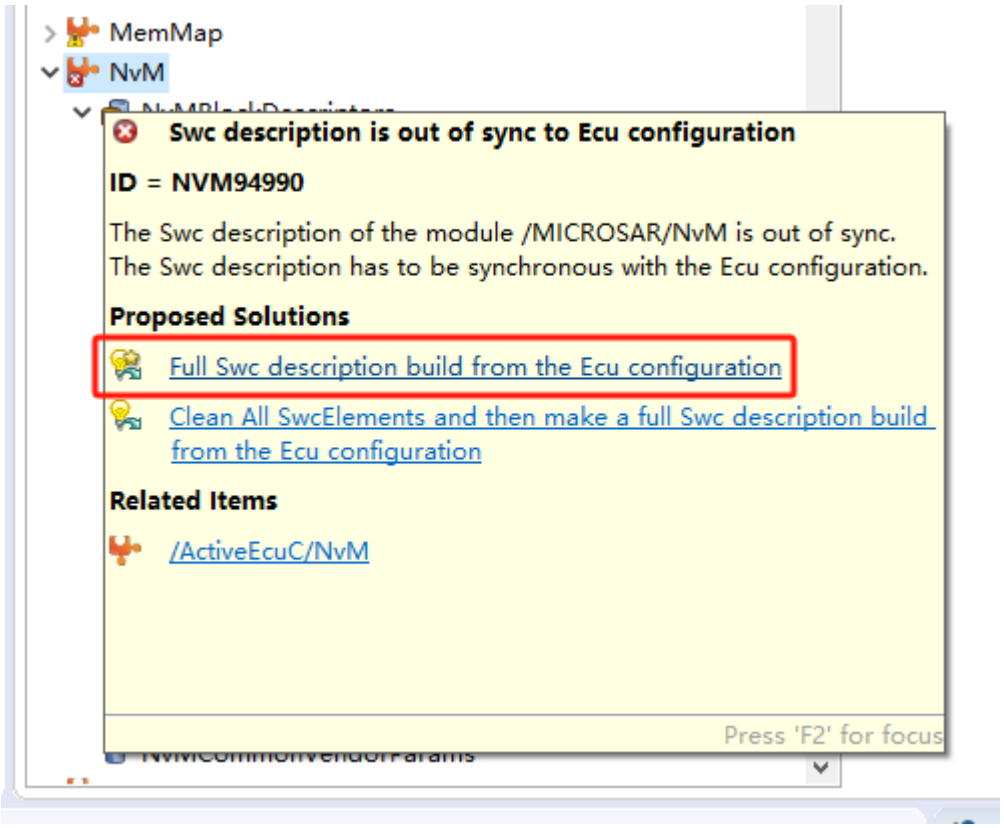



## 3.2、NvM 选择对应的FeeBlock

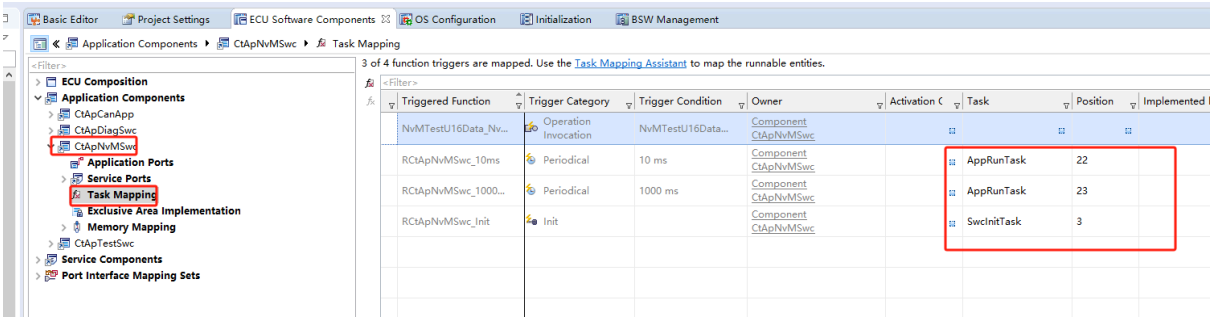
NvM新增的Block 选择FeeRef



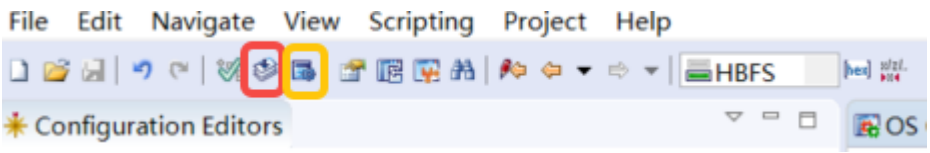



# 3.3、CtApNvMSwc的Task映射

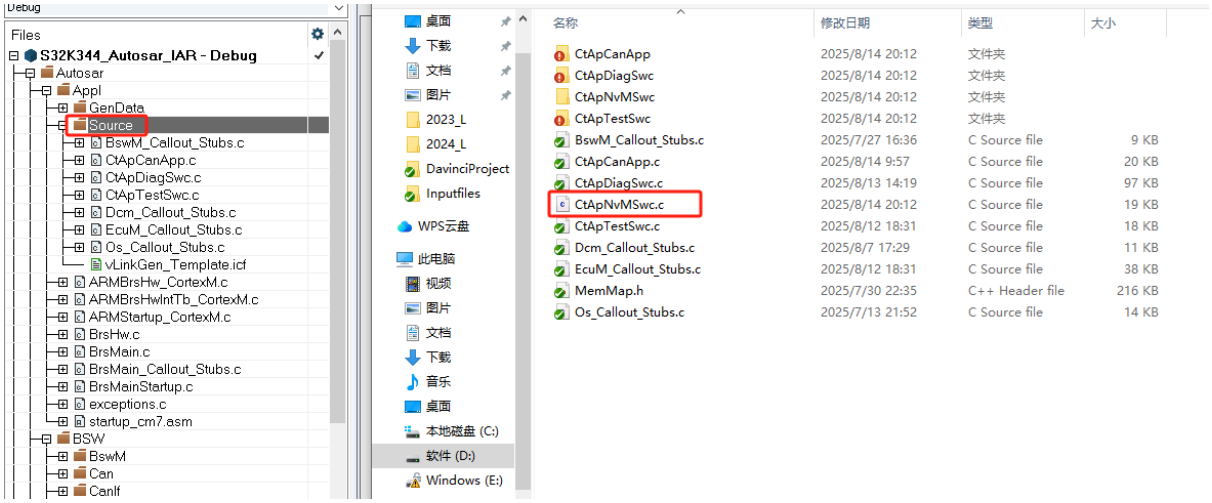


最后生成所有模块， 然后生成Swc代码

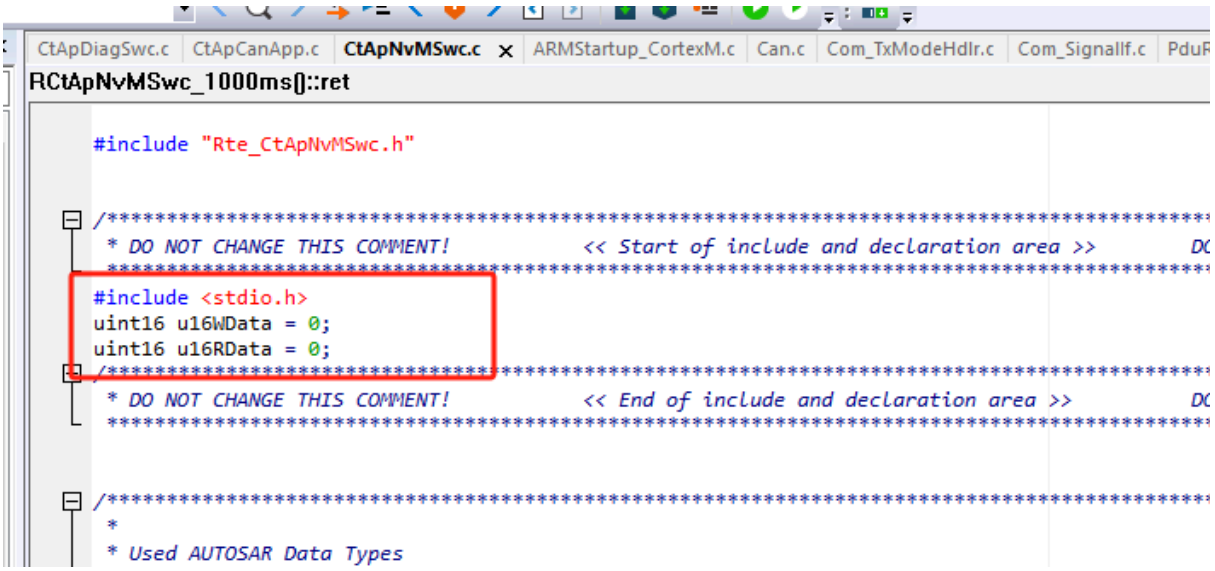



# 四、代码验证

IAR 添加 CtApNvMSwc.c 到工程




添加测试代码

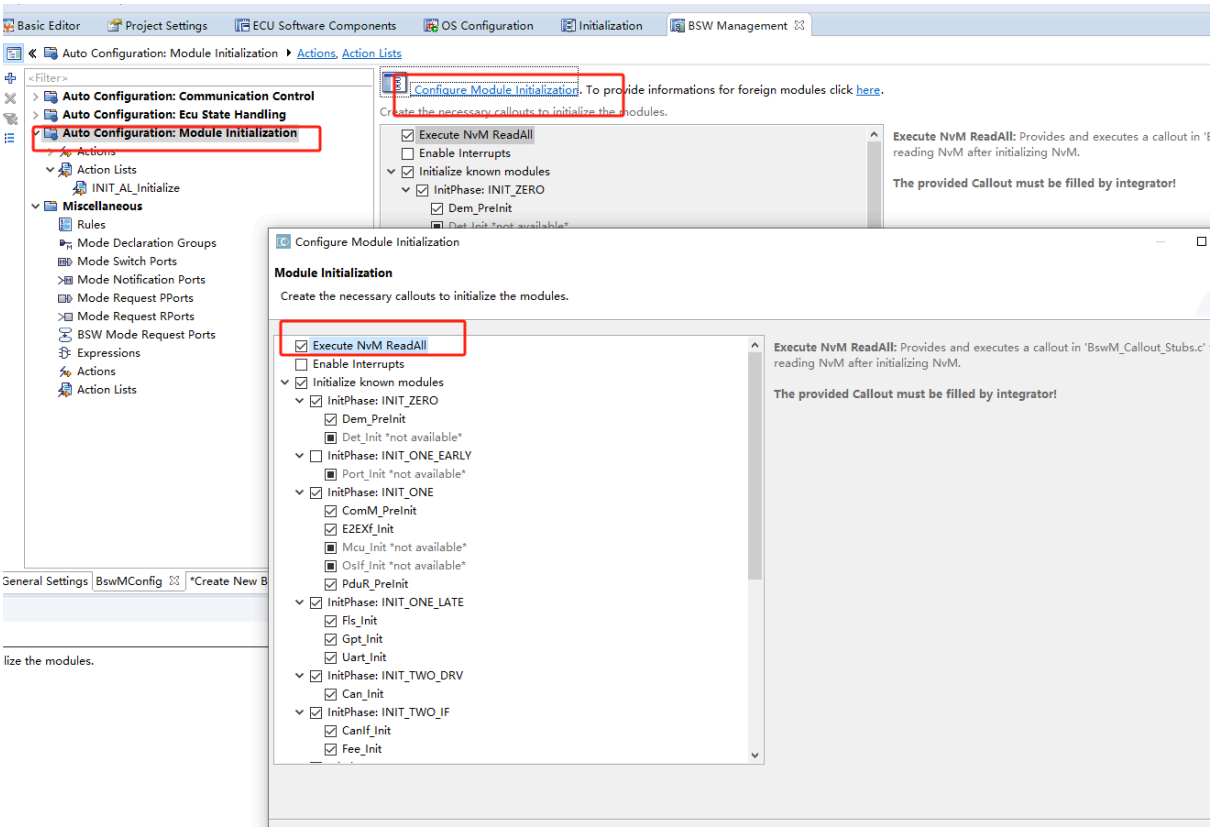


```
**********************************************************************************************************/

FUNC(void, CtApNvMSwc_CODE) RCtApNvMSwc_1000ms(void) /* PRQA S 0624, 3206 */ /* MD_Rte_0624, MD_Rte_3206 */
{
/**********************************************************************************************************
 * DO NOT CHANGE THIS COMMENT!          << Start of runnable implementation >>            DO NOT CHANGE THIS COMMENT!
 * Symbol: RCtApNvMSwc_1000ms
 **********************************************************************************************************/
  Std_ReturnType ret = 0;
  NvM_RequestResultType  ErrorStatus = 0;

  ret = Rte_Call_NvMTestU16Data_NvMService_AC3_GetErrorStatus(&ErrorStatus);
  if((ret == E_OK) && (ErrorStatus != NVM_REQ_PENDING) ){

    static uint8 IsWrite = 0;
    if(0 == IsWrite){
      ret = Rte_Call_NvMTestU16Data_NvMService_AC3_ReadBlock(&u16RData);
    }else{
      u16WData++;// = 0x2233;
      ret = Rte_Call_NvMTestU16Data_NvMService_AC3_WriteBlock(&u16WData);
    }
    IsWrite = !IsWrite;
    printf("ret=%d, IsWrite=%d, %s Data: 0x%04x\r\n",ret, !IsWrite, !IsWrite?"Write":"Read", IsWrite?u16RData:u16WData);
  }

/**********************************************************************************************************
 * DO NOT CHANGE THIS COMMENT!          << End of runnable implementation >>              DO NOT CHANGE THIS COMMENT!
 **********************************************************************************************************/
```

可以看见读写正常，因为回调出来之后数据才会更新，所以读会比写慢，跟Fls/Fee/Nvm的顺序和运行周期也有关联，我们后面调整

```
Hello World, 238
Hello World, 239
Hello World, 240
ret=0, IsWrite=0, Read Data: 0x001d
Hello World, 241
Hello World, 242
Hello World, 243
Hello World, 244
ret=0, IsWrite=1, Write Data: 0x001f
Hello World, 245
Hello World, 246
Hello World, 247
Hello World, 248
ret=0, IsWrite=0, Read Data: 0x001e
Hello World, 249
Hello World, 250
Hello World, 251
Hello World, 252
ret=0, IsWrite=1, Write Data: 0x0020
Hello World, 253
Hello World, 254
Hello World, 255
Hello World, 256
ret=0, IsWrite=0, Read Data: 0x001f
Hello World, 257
```

# 五、添加BswM Init时的NvM_ReadAll




## 生成BswM

## 然后添加Read All的代码

```
 *  Additional configured User includes
 ***********************************************************************************************************/

/***********************************************************************************************************
 * DO NOT CHANGE THIS COMMENT!          <USERBLOCK User Includes>                    DO NOT CHANGE THIS COMMENT!
 ***********************************************************************************************************/
#include "Fls.h"
/***********************************************************************************************************
 * DO NOT CHANGE THIS COMMENT!          </USERBLOCK>                                 DO NOT CHANGE THIS COMMENT!
 ***********************************************************************************************************/


/***********************************************************************************************************
 *  CALLOUT FUNCTIONS
 ***********************************************************************************************************/
#define BSWM_START_SEC_CODE
#include "BswM_MemMap.h" /* PRQA S 5087 */ /* MD_MSR_MemMap */

/***********************************************************************************************************
 *  GENERIC CALLOUTS
 ***********************************************************************************************************/
FUNC(void, BSWM_CODE) BswM_INIT_NvMReadAll(void)
{
/***********************************************************************************************************
 * DO NOT CHANGE THIS COMMENT!          <USERBLOCK BswM_INIT_NvMReadAll>             DO NOT CHANGE THIS COMMENT!
 ***********************************************************************************************************/
  {
      NvM_RequestResultType Int_NvMultiBlockStatus;
      NvM_ReadAll();
      do
      {
          NvM_MainFunction();
          Fee_MainFunction();
          Fls_MainFunction();
          NvM_GetErrorStatus(NvMConf___MultiBlockRequest, &Int_NvMultiBlockStatus);

#if defined (VGEN_ENABLE_DRVWD)
          Wdg_Trigger();
#endif /* VGEN_ENABLE_DRVWD */
      }
      while (Int_NvMultiBlockStatus == NVM_REQ_PENDING);
  }

/***********************************************************************************************************
```

初始化时这里调用完之后Rte.c 中的NvM 对应Block 的Ram数据就会读出来了的

```
#define RTE_START_SEC_CONST_DEFAULT_RTE_CDATA_GROUP_UNSPECIFIED
#include "Rte_MemMap.h" /* PRQA S 5087 */ /* MD_MSR_MemMap */

/* PRQA S 3408, 1514, 1533 L1 */ /* MD_Rte_3408, MD_Rte_1514, MD_Rte_1533 */
CONST(uint16, RTE_CONST_DEFAULT_RTE_CDATA_GROUP) Rte_CtApNvMSwc_UInt16InitValue = 0U;
/* PRQA L:L1 */

#define RTE_STOP_SEC_CONST_DEFAULT_RTE_CDATA_GROUP_UNSPECIFIED
#include "Rte_MemMap.h" /* PRQA S 5087 */ /* MD_MSR_MemMap */


/**********************************************************************************************************************
 * Per-Instance Memory
 **********************************************************************************************************************/

#define RTE_START_SEC_VAR_DEFAULT_RTE_PIM_GROUP_UNSPECIFIED
#include "Rte_MemMap.h" /* PRQA S 5087 */ /* MD_MSR_MemMap */

VAR(uint16, RTE_VAR_DEFAULT_RTE_PIM_GROUP) Rte_CtApNvMSwc_NvMTestU16Data; /* PRQA S 3408, 1504, 1514, 1533 */ /* MD_Rte_3408, MD_MSR_Rule8.7, MD_Rte_1514, MD_Rte_1533 */

#define RTE_STOP_SEC_VAR_DEFAULT_RTE_PIM_GROUP_UNSPECIFIED
#include "Rte_MemMap.h" /* PRQA S 5087 */ /* MD_MSR_MemMap */


/**********************************************************************************************************************
 * TxAck/ModeSwitchAck Flags
 **********************************************************************************************************************/
```